RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 4 (Zeta)
Topic(s): Calculus II
(Integration, Applications, DE) (Solutions)

Source of Question: TJC Prelim 9758/2022/01/Q1

1 Find

(a) $\int \sin^{-1} 2x \, \mathrm{d}x$, [3]

(b) $\int \frac{1}{x^2 - nx + n^2} \, \mathrm{d}x$, where n is a non-zero constant. [2]

Solution:

(a)

$$\int \sin^{-1} 2x \, \mathrm{d}x = \int 1.\sin^{-1} 2x \, \mathrm{d}x$$

$u = \sin^{-1} 2x$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2$ $\quad v = x$

$$= x\sin^{-1} 2x - \int x.\frac{2}{\sqrt{1-4x^2}} \, \mathrm{d}x$$

$$= x\sin^{-1} 2x + \frac{1}{4}\int (-8x)\left(1-4x^2\right)^{-\frac{1}{2}} \, \mathrm{d}x$$

$$= x\sin^{-1} 2x + \frac{1}{4}.\frac{\left(1-4x^2\right)^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= x\sin^{-1} 2x + \frac{1}{2}\sqrt{1-4x^2} + c$$

(b)

$$\int \frac{1}{x^2 - nx + n^2} \, \mathrm{d}x = \int \frac{1}{\left(x - \frac{1}{2}n\right)^2 - \frac{1}{4}n^2 + n^2} \, \mathrm{d}x$$

$$= \int \frac{1}{\left(x - \frac{1}{2}n\right)^2 + \frac{3}{4}n^2} \, \mathrm{d}x$$

$$= \frac{2}{\sqrt{3}n} \tan^{-1}\left(\frac{x - \frac{1}{2}n}{\frac{\sqrt{3}}{2}n}\right) + c$$

$$= \frac{2}{\sqrt{3}n} \tan^{-1}\left(\frac{2x - n}{\sqrt{3}n}\right) + c$$

Source of Question: TMJC Prelim 9758/2022/01/Q2(modified)

2 **(a)** Find $\displaystyle\int \frac{1}{\sqrt{\left(1-x^2\right)\sin^{-1}x}}\,\mathrm{d}x$. [2]

(b) Find $\displaystyle\int \frac{x-3}{x^2+2x-3}\,\mathrm{d}x$. [3]

Solution:

(a)

$$\int \frac{1}{\sqrt{\left(1-x^2\right)\sin^{-1}x}}\,\mathrm{d}x = \int \frac{1}{\sqrt{\left(1-x^2\right)}}\left(\sin^{-1}x\right)^{-\frac{1}{2}}\,\mathrm{d}x$$

$$= \frac{\left(\sin^{-1}x\right)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 2\sqrt{\sin^{-1}x} + C$$

The integrand is of the form $\mathrm{f}'(x)(\mathrm{f}(x))^n$, where $\mathrm{f}(x) = \sin^{-1}x$ and $n = -\frac{1}{2}$.

(b)

$$\int \frac{x-3}{x^2+2x-3}\,\mathrm{d}x = \int \frac{x-3}{(x-1)(x+3)}\,\mathrm{d}x$$

$$= -\frac{1}{2}\int \frac{1}{x-1}\,\mathrm{d}x + \frac{3}{2}\int \frac{1}{x+3}\,\mathrm{d}x$$

$$= -\frac{1}{2}\ln|x-1| + \frac{3}{2}\ln|x+3| + C$$

Alternatively,

$$\int \frac{x-3}{x^2+2x-3}\,\mathrm{d}x = \frac{1}{2}\int \frac{2x+2}{x^2+2x-3}\,\mathrm{d}x - \int \frac{4}{x^2+2x-3}\,\mathrm{d}x$$

$$= \frac{1}{2}\ln\left|x^2+2x-3\right| - 4\int \frac{1}{(x+1)^2-4}\,\mathrm{d}x$$

$$= \frac{1}{2}\ln|(x+3)(x-1)| - \frac{4}{4}\ln\left|\frac{x+1-2}{x+1+2}\right| + C$$

$$= \frac{1}{2}\ln|x+3| + \frac{1}{2}\ln|x-1| - \left(\ln|x-1| - \ln|x+3|\right) + C$$

$$= -\frac{1}{2}\ln|x-1| + \frac{3}{2}\ln|x+3| + C$$

Source of Question: NYJC Prelim 9758/2022/02/Q5 (modified)

3 **(a)** **(i)** Differentiate $\sqrt{x^2+4}$ with respect to x. [1]

(ii) Hence find $\int \frac{x^3}{\sqrt{x^2+4}}\,\mathrm{d}x$. [3]

(b) Find **(i)** $\int \sin^3 mx\,\mathrm{d}x$, where m is a non-zero constant, [2]

(ii) $\int \cos^4 3x\mathrm{d}x$. [4]

Solution:

(a)(i)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{x^2+4}\right)=\frac{2x}{2\sqrt{x^2+4}}=\frac{x}{\sqrt{x^2+4}}$$

(a)(ii)

$$u=x^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x}=\frac{x}{\sqrt{x^2+4}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x}=2x \qquad v=\sqrt{x^2+4}$$

$$\int \frac{x^3}{\sqrt{x^2+4}}\mathrm{d}x = x^2\sqrt{x^2+4}-\int 2x\sqrt{x^2+4}\,\mathrm{d}x$$

$$= x^2\sqrt{x^2+4}-\frac{2\left(x^2+4\right)^{\frac{3}{2}}}{3}+C$$

(b)(i)

$$\int \sin^3 mx\,\mathrm{d}x = \int \sin mx\left(\sin^2 mx\right)\mathrm{d}x$$

$$= \int \sin mx\left(1-\cos^2 mx\right)\mathrm{d}x$$

$$= \int \sin mx - \sin mx\cos^2 mx\,\mathrm{d}x$$

$$= -\frac{\cos mx}{m}+\frac{\cos^3 mx}{3m}+C$$

(b)(ii)

$$\int \cos^4 3x\,\mathrm{d}x = \frac{1}{4}\int \left(1+\cos 6x\right)^2\mathrm{d}x$$

$$= \int \frac{1}{4}+\frac{\cos 6x}{2}+\frac{1}{4}\cos^2 6x\,\mathrm{d}x$$

$$= \int \frac{1}{4}+\frac{\cos 6x}{2}+\frac{1}{4}\left(\frac{1+\cos 12x}{2}\right)\mathrm{d}x$$

$$= \frac{1}{4}x+\frac{\sin 6x}{12}+\frac{1}{8}x+\frac{\sin 12x}{96}+C$$

$$= \frac{3}{8}x+\frac{\sin 6x}{12}+\frac{\sin 12x}{96}+C$$

Source of Question: NJC Prelim 9758/2022/01/Q1

4 **(i)** Given that $I_n = \int_1^{\mathrm{e}} x(\ln x)^n \,\mathrm{d}x$ for $n \in \mathbb{Z},\ n \geq 0,$ show that

$$I_n = \frac{\mathrm{e}^2}{2} - \frac{n}{2} I_{n-1}$$

for all $n \in \mathbb{Z}^+$. [2]

(ii) Find the exact volume of the solid generated when the region bounded by the curve $y = \sqrt{x}\ln x,$ the x-axis and the line $x = \mathrm{e}$ is rotated completely about the x-axis. [3]

Solution:

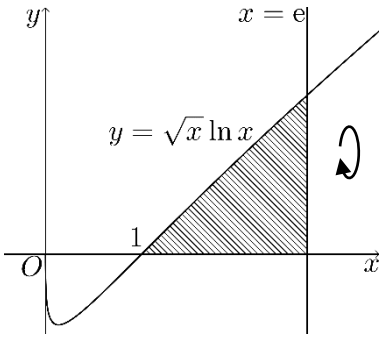
(i)

$$\begin{aligned} I_n &= \int_1^{\mathrm{e}} x(\ln x)^n \,\mathrm{d}x \\ &= \left[\frac{x^2}{2}(\ln x)^n\right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} \frac{x^2}{2}\cdot n(\ln x)^{n-1}\frac{1}{x}\,\mathrm{d}x \\ &= \frac{\mathrm{e}^2}{2} - \frac{n}{2}\int_1^{\mathrm{e}} x(\ln x)^{n-1}\,\mathrm{d}x \\ &= \frac{\mathrm{e}^2}{2} - \frac{n}{2} I_{n-1} \text{ (shown)} \end{aligned}$$

(ii)

Volume

$$\begin{aligned} &= \pi\int_1^{\mathrm{e}} x(\ln x)^2 \,\mathrm{d}x \\ &= \pi I_2 \\ &= \pi\left(\frac{\mathrm{e}^2}{2} - \frac{2}{2} I_1\right) \\ &= \pi\left[\frac{\mathrm{e}^2}{2} - \left(\frac{\mathrm{e}^2}{2} - \frac{1}{2} I_0\right)\right] \\ &= \frac{\pi}{2} I_0 = \frac{\pi}{2}\left[\frac{x^2}{2}\right]_1^{\mathrm{e}} \\ &= \frac{\pi}{4}\left(\mathrm{e}^2 - 1\right) \text{ units}^3 \end{aligned}$$



Alternatively:

Volume

$$= \pi \int_1^{\mathrm{e}} x(\ln x)^2 \, \mathrm{d}x$$

$$= \pi \left\{ \left[\frac{x^2}{2}(\ln x)^2 \right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} \frac{x^2}{2} \cdot 2(\ln x) \frac{1}{x} \mathrm{d}x \right\}$$

$$= \pi \left[\frac{\mathrm{e}^2}{2} - \int_1^{\mathrm{e}} x(\ln x) \mathrm{d}x \right]$$

$$= \pi \left\{ \frac{\mathrm{e}^2}{2} - \left[\frac{x^2}{2}(\ln x) \right]_1^{\mathrm{e}} + \int_1^{\mathrm{e}} \frac{x^2}{2} \cdot \frac{1}{x} \mathrm{d}x \right\}$$

$$= \pi \left\{ \frac{\mathrm{e}^2}{2} - \frac{\mathrm{e}^2}{2} + \left[\frac{x^2}{4} \right]_1^{\mathrm{e}} \right\}$$

$$= \frac{\pi}{4}\left(\mathrm{e}^2 - 1\right) \text{ units}^3$$

$u = (\ln x)^2$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2(\ln x)\frac{1}{x}$ $\quad v = \frac{x^2}{2}$

$u = \ln x$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ $\quad v = \frac{x^2}{2}$

Source of Question: ACJC Prelim 9758/2022/01/Q4 (modified)

5 (a) The continuous function $\mathrm{f}(x)$, where $\mathrm{f}(x) > 0$, is strictly decreasing for $x \geq 1$.
Sketch the curve $y = \mathrm{f}(x)$ for $k < x < k+1$ where k is an integer and $k \geq 1$.
By comparing the areas of appropriate rectangles and the area under the curve $y = \mathrm{f}(x)$, show that for any integer $k \geq 1$,

$$\mathrm{f}(k+1) < \int_k^{k+1} \mathrm{f}(x)\,\mathrm{d}x < \mathrm{f}(k).$$ [2]

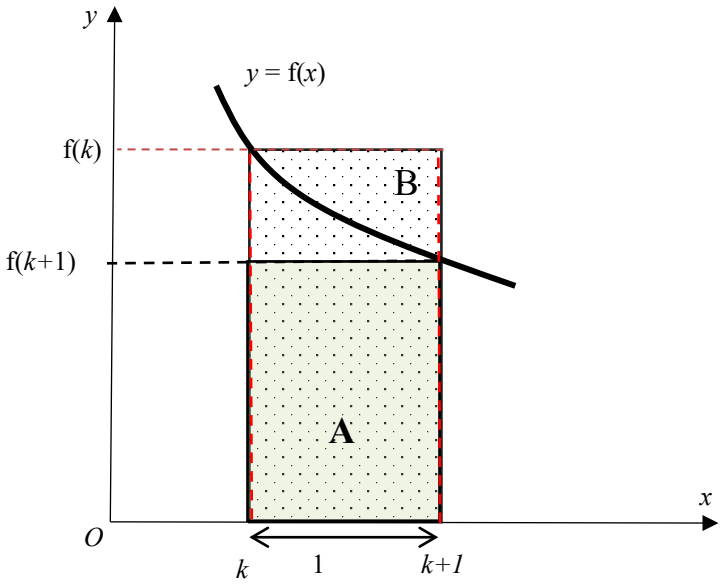
(b) Hence show that

(i) $\ln 10 < 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{9} < 1 + \ln 9$. [3]

(ii) $\ln\left((n-1)!\right) < n\ln n - n + 1 < \ln\left(n!\right)$ for $n > 1$. [4]

Solution:

(a)



Area under curve = $\int_k^{k+1} \mathrm{f}(x)\,\mathrm{d}x$

Area of rectangle A= $\mathrm{f}(k+1)$ x 1

Area of rectangle B= $\mathrm{f}(k)$ x 1

As seen from diagram:

$\mathrm{f}(k+1) < \int_k^{k+1} \mathrm{f}(x)\,\mathrm{d}x < \mathrm{f}(k)$

(b)(i)

Let $\mathrm{f}(x)=\dfrac{1}{x}$.

$$\int_1^{10}\frac{1}{x}\mathrm{d}x=\int_1^{2}\frac{1}{x}\mathrm{d}x+\int_2^{3}\frac{1}{x}\mathrm{d}x+\int_3^{4}\frac{1}{x}\mathrm{d}x+....+\int_9^{10}\frac{1}{x}\mathrm{d}x$$

$$<\mathrm{f}(1)+\mathrm{f}(2)+\mathrm{f}(3)+.....+\mathrm{f}(9)$$

$$=\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{9}$$

$$\int_1^{9}\frac{1}{x}\mathrm{d}x=\int_1^{2}\frac{1}{x}\mathrm{d}x+\int_2^{3}\frac{1}{x}\mathrm{d}x+\int_3^{4}\frac{1}{x}\mathrm{d}x+....+\int_8^{9}\frac{1}{x}\mathrm{d}x$$

$$>\mathrm{f}(2)+\mathrm{f}(3)+\mathrm{f}(4)+.....+\mathrm{f}(9)$$

$$=\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{9}=\sum_{k=1}^{9}\frac{1}{k}-1$$

$$\Rightarrow 1+\int_1^{9}\frac{1}{x}\mathrm{d}x>1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{9}$$

Combining both inequalities we have

$$\int_1^{10}\frac{1}{x}\mathrm{d}x<\sum_{k=1}^{9}\frac{1}{k}<1+\int_1^{9}\frac{1}{x}\mathrm{d}x$$

$$\left[\ln x\right]_1^{10}<1+\frac{1}{2}+\frac{1}{3}+.....\frac{1}{9}<1+\left[\ln x\right]_1^{9}$$

$$\ln 10<1+\frac{1}{2}+\frac{1}{3}+.....\frac{1}{9}<1+\ln 9$$

(b)(ii)

We consider the function f(x) = ln x. This function is strictly increasing, so the inequality in (a) changes to $\mathrm{f}\left(k\right)<\int_k^{k+1}\mathrm{f}\left(x\right)\mathrm{d}x<\mathrm{f}\left(k+1\right)$.

We thus have

$$\sum_{k=1}^{n-1}\mathrm{f}\left(k\right)<\sum_{k=1}^{n-1}\int_k^{k+1}\mathrm{f}\left(x\right)\mathrm{d}x<\sum_{k=1}^{n-1}\mathrm{f}\left(k+1\right)$$

$$\Rightarrow \ln 1+\ln 2+...+\ln(n-1)<\int_1^{2}\mathrm{f}\left(x\right)\mathrm{d}x+...+\int_{n-1}^{n}\mathrm{f}\left(x\right)\mathrm{d}x<\ln 2+...+\ln n$$

$$\Rightarrow \ln\left((n-1)!\right)<\int_1^{n}\ln x\,\mathrm{d}x<\ln\left(n!\right)$$

$$\Rightarrow \ln\left((n-1)!\right)<\left[x\ln x-x\right]_1^{n}<\ln\left(n!\right)$$

$$\Rightarrow \ln\left((n-1)!\right)<n\ln n-n+1<\ln\left(n!\right)$$

Source of Question: VJC Prelim 9758/2022/01/Q9

6 The curve C has equation $\frac{1}{3}x^2 + y^2 - 2y = 0$.

(i) Sketch C. [2]

(ii) Use the substitution $x = p\sin\theta$ to show that

$$\int_0^p \sqrt{p^2 - x^2}\,\mathrm{d}x = \frac{p^2\pi}{4},$$

where p is a positive constant. [4]

(iii) The region R is bounded by C, the line $x = \sqrt{3}$ and the x-axis.

Find the exact area of R. [3]

(iv) R is rotated completely about the y-axis.

Find the exact volume of the solid obtained. [3]

(v) Describe a pair of transformations which transforms the graph of C onto the graph of $x^2 + y^2 = 1$. [2]

Solution:

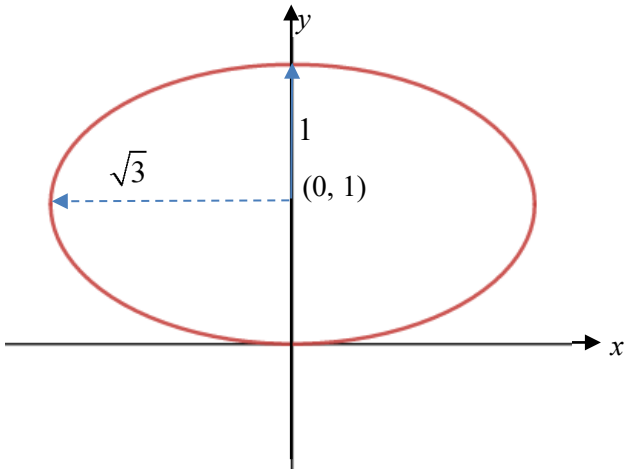
(i)

$\frac{1}{3}x^2 + y^2 - 2y = 0$

$\frac{1}{3}x^2 + (y-1)^2 - 1 = 0$

$\frac{1}{3}x^2 + (y-1)^2 = 1$

This is an ellipse centered at (0, 1) with semi-minor axis 1 and semi-major axis $\sqrt{3}$.



(ii)

$$\int_0^p \sqrt{p^2-x^2}\ \mathrm{d}x$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{p^2-p^2\sin^2\theta}\,(p\cos\theta)\,\mathrm{d}\theta$$

$$= p^2\int_0^{\frac{\pi}{2}} \sqrt{\cos^2\theta}\,(\cos\theta)\,\mathrm{d}\theta$$

$$= p^2\int_0^{\frac{\pi}{2}} \cos^2\theta\,\mathrm{d}\theta$$

$$= p^2\int_0^{\frac{\pi}{2}} \left(\frac{\cos 2\theta+1}{2}\right)\mathrm{d}\theta$$

$$= p^2\left[\frac{\sin 2\theta}{4}+\frac{\theta}{2}\right]_0^{\frac{\pi}{2}}$$

$$= p^2\left(\frac{\sin 2\left(\frac{\pi}{2}\right)}{4}+\frac{\frac{\pi}{2}}{2}\right) = \frac{p^2\pi}{4}$$

$x = p\sin\theta$

$\mathrm{d}x = p\cos\theta\,\mathrm{d}\theta$

When $x=0$, $\theta=0$

When $x=p$, $\theta=\frac{\pi}{2}$

(iii)

$$\frac{1}{3}x^2+(y-1)^2=1$$

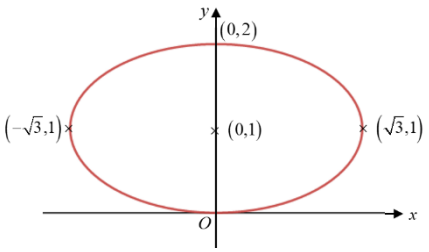
$$y = 1-\sqrt{1-\frac{1}{3}x^2} \quad (\because y<1)$$

$$\text{Area} = \int_0^{\sqrt{3}}\left(1-\sqrt{1-\frac{1}{3}x^2}\right)\mathrm{d}x$$

$$= \int_0^{\sqrt{3}} 1\,\mathrm{d}x - \frac{1}{\sqrt{3}}\int_0^{\sqrt{3}}\sqrt{3-x^2}\ \mathrm{d}x$$

$$= \sqrt{3}-\frac{1}{\sqrt{3}}\left(\frac{3\pi}{4}\right) \quad \text{using (ii)}$$

$$= \sqrt{3}-\frac{\sqrt{3}\pi}{4}$$



(iv)

$$\text{Volume} = \pi\left(\sqrt{3}\right)^2(1)-\pi\int_0^1 x^2\ \mathrm{d}y$$

$$= 3\pi-\pi\int_0^1\left(6y-3y^2\right)\mathrm{d}y$$

$$= 3\pi-\pi\left[3y^2-y^3\right]_0^1$$

$$= \pi$$

(v)

Translate 1 units in the negative y-direction.

Stretch the resultant curve by a factor of $\frac{1}{\sqrt{3}}$ parallel to the x-axis.

Source of Question: CJC Prelim 9758/2022/02/Q5

7 A curve C_1 has equation $y = \mathrm{f}(x)$, where $k > 1$ and

$$\mathrm{f}(x) = \begin{cases} x + k & \text{for } -k < x \le 0, \\ (k+1) - \dfrac{(2k-x)^2}{4k^2} & \text{for } 0 < x \le 2k, \\ 0 & \text{otherwise.} \end{cases}$$

(i) Sketch the curve C_1. [2]

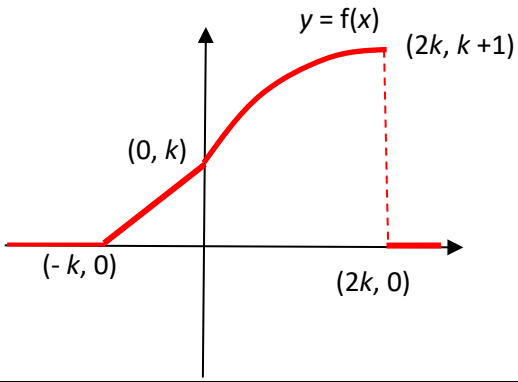
(ii) The region R is bounded by the curve C_1, the line $x = 1$ and the x-axis. Find the exact area of R, leaving your answer in terms of k. [6]

(iii) On the same diagram on part **(i)**, sketch the curve C_2 with equation $y = x^2 \tan^{-1} x$. [1]

(iv) The region S is bounded by the curves C_1 and C_2, and the x-axis. Given that $k = 3$, determine the volume of solid of revolution formed when S is rotated 2π radians about the x-axis, leaving your answer to 2 decimal places. [3]

Solution:

(i)



(ii)

Required Area

$$= \int_{-k}^{0} (x+k)\,dx + \int_{0}^{1} (k+1) - \frac{(2k-x)^2}{4k^2}\,dx$$

$$= \left[\frac{x^2}{2} + kx\right]_{-k}^{0} + \left[(k+1)x - \frac{(2k-x)^3}{4k^2(3)(-1)}\right]_{0}^{1}$$

$$= 0 - \left[\frac{k^2}{2} - k^2\right] + \left[(k+1) - \frac{(2k-1)^3}{4k^2(3)(-1)}\right] - \left[-\frac{(2k)^3}{4k^2(3)(-1)}\right]$$

$$= \frac{k^2}{2} + \frac{k}{3} + 1 + \frac{(2k-1)^3}{12k^2}$$

OR

Required Area

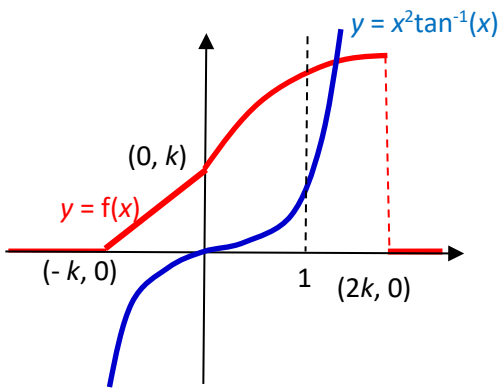
$$= \frac{1}{2}(k)(k) + \int_{0}^{1} (k+1) - \frac{(2k-x)^2}{4k^2}\,dx$$

$$= \frac{k^2}{2} + \left[(k+1)x - \frac{(2k-x)^3}{4k^2(3)(-1)}\right]_{0}^{1}$$

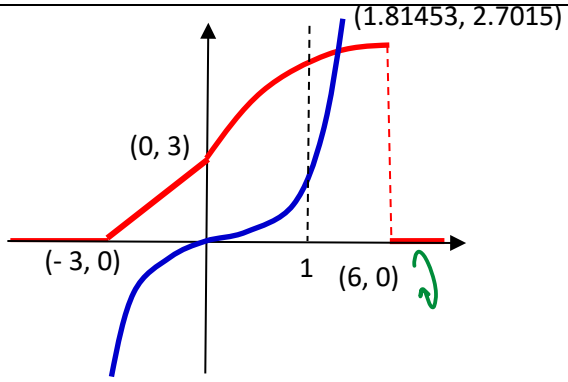
$$= \frac{k^2}{2} + \left[(k+1) - \frac{(2k-1)^3}{4k^2(3)(-1)} \right] - \left[-\frac{(2k)^3}{4k^2(3)(-1)} \right]$$

$$= \frac{k^2}{2} + \frac{k}{3} + 1 + \frac{(2k-1)^3}{12k^2}$$

(iii)



(iv)



Point of intersection between C_1 and C_2 is at $x = 1.81453$

Volume of S

$$= \pi \left\{ \int_{-3}^{0} (x+3)^2 dx + \int_{0}^{1.81453} \left[(3+1) - \frac{(2(3)-x)^2}{4(3)^2} \right]^2 dx - \int_{0}^{1.81453} \left[x^2 \tan^{-1}(x) \right]^2 dx \right\}$$

= 77.54 (2 d.p.) unit^3

OR

Volume of S

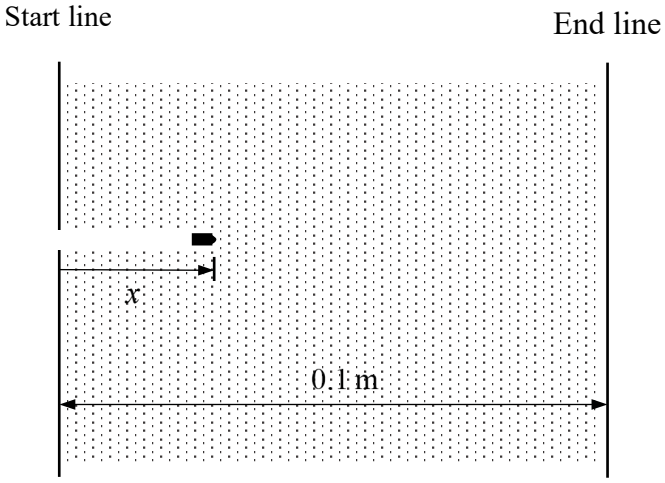
$$= \frac{1}{3}\pi(3)^2(3) + \pi \left\{ \int_{0}^{1.81453} \left[(3+1) - \frac{(2(3)-x)^2}{4(3)^2} \right]^2 dx - \int_{0}^{1.81453} \left[x^2 \tan^{-1}(x) \right]^2 dx \right\}$$

= 77.54 (2 d.p.) unit^3

Source of Question: DHS Prelim 9758/2021/01/Q11

8 A company conducts experiments to investigate the resistive power of armoured plates of different materials against a bullet fired from a rifle. The plates used for the experiment have a consistent thickness of $0.1\,\text{m}$. Once the plates are positioned, a bullet is fired at the start line of the plate with a fixed velocity of $100\,\text{m s}^{-1}$.

For a duration of t s after it is fired, the bullet is x m away from the start line and travels at a velocity $v\,\text{m s}^{-1}$. The setup is shown below.



(i) When a plate of material P is used, v and x are modelled by the following differential equations:

$$\text{(A)} \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = v,$$

$$\text{(B)} \qquad \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \beta\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = 0,$$

where β is a positive constant.

(a) By substituting equation (A), show that equation (B) can be written as $\dfrac{\mathrm{d}v}{\mathrm{d}t} = -\beta v^2$. [1]

(b) Find v in terms of β and t and hence find x in terms of β and t. [5]

(c) Given that it is found that the bullet exits the end line of the plate with a velocity of $80\,\text{m s}^{-1}$, find the time taken for the bullet to pass through the plate of material P, leaving your answer to 5 decimal places. [3]

(ii) The company develops another plate made of a new material Q. When the experiment is conducted with a plate of material Q, it is found that the bullet is slowed down in such a way that $\dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}$ is inversely proportional to $\mathrm{e}^{10\,000t}$. It is given that $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 100$ and

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -10^6$ when $t = 0$. Write down a differential equation to model the bullet as it travels through the plate of material *Q*. Solve this differentiation equation to find x in terms of t for this model. [4]

(iii) The company claims that material *Q* provides better protection against bullets than material *P*. Explain, with justification, whether the company's claim is valid. [1]

Solution:

(i)(a)

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \beta\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = 0$$

Let $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, $\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$, we have

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \beta v^2 = 0$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -\beta v^2$$

(i)(b)

$$v^{-2}\frac{\mathrm{d}v}{\mathrm{d}t} = -\beta$$

$$\int v^{-2}\,\mathrm{d}v = -\beta\int 1\,\mathrm{d}t$$

$$-\frac{1}{v} = -\beta t + C$$

When $t = 0$, $v = 100$, $C = -\frac{1}{100}$.

$$-\frac{1}{v} = -\beta t - \frac{1}{100}$$

$$\frac{1}{v} = \beta t + \frac{1}{100}$$

$$v = \frac{1}{\beta t + 0.01} \quad \text{--- (1)}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{\beta t + 0.01}$$

$$x = \int \frac{1}{\beta t + 0.01}\,\mathrm{d}t$$

$$x = \frac{1}{\beta}\ln\left(\beta t + 0.01\right) + D \quad (\text{since } \beta > 0)$$

When $t = 0$, $x = 0$,

$$0 = \frac{1}{\beta}\ln\left(0.01\right) + D$$

$$D = -\frac{1}{\beta}\ln\left(0.01\right)$$

	$x = \frac{1}{\beta}\ln(\beta t + 0.01) - \frac{1}{\beta}\ln(0.01)$ $x = \frac{1}{\beta}\left[\ln\left(\frac{\beta t + 0.01}{0.01}\right)\right]$ $x = \frac{1}{\beta}\ln(100\beta t + 1)$ --- (2)
(i)(c)	Let $t = T$ when the bullet leaves plate of material X. When $t = T$, $v = 80$, from (1), $80 = \frac{1}{\beta T + 0.01}$ $\beta T + 0.01 = \frac{1}{80}$ $\beta T = \frac{1}{400} = 0.0025$ $\beta = \frac{1}{400T}$ When $t = T$, $x = 0.1$ and $\beta = \frac{1}{400T}$, from (2), $0.1 = \frac{1}{\left(\frac{1}{400T}\right)}\ln\left(100\left(\frac{1}{400T}\right)T + 1\right)$ $0.1 = 400T\ln\left(\frac{5}{4}\right)$ $T = \frac{0.1}{400\ln(1.25)}$ $= 0.00112035502$ $= 0.00112$ seconds
(ii)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{k}{\mathrm{e}^{10000t}} = k\mathrm{e}^{-10^4 t}$ When $t = 0$, $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -10^6$, $-10^6 = k\mathrm{e}^{-10^4(0)}$ $k = -10^6$ Hence, $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -10^6\mathrm{e}^{-10^4 t}$ $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{-10^6}{-10^4}\mathrm{e}^{-10^4 t} + E$ $= 100\mathrm{e}^{-10^4 t} + E$

	When $t = 0$, $\frac{\mathrm{d}x}{\mathrm{d}t} = 100$, $100 = 100\mathrm{e}^{-10^4(0)} + E$ $E = 0$ $\frac{\mathrm{d}x}{\mathrm{d}t} = 100\mathrm{e}^{-10^4 t}$ --- (3) $x = -\frac{100}{10^4}\mathrm{e}^{-10^4 t} + F$ $x = -0.01\mathrm{e}^{-10^4 t} + F$ When $t = 0, x = 0$, $0 = -0.01\mathrm{e}^{-10^4(0)} + F$ $F = 0.01$ $x = 0.01 - 0.01\mathrm{e}^{-10^4 t}$ $x = 0.01\left(1 - \mathrm{e}^{-10^4 t}\right)$ --- (4)
(iii)	**Method 1** When $t = 0.0011204$, from (4), $x = 0.01\left(1 - \mathrm{e}^{-10^4(0.0011204)}\right)$ $\approx 0.001 < 0.1$ In the same amount of time it takes for the bullet to penetrate the entire plate of material P of 0.1m, it only travels 0.01m through plate of material Q. Hence, the company's claim is valid. **Method 2** From (4), as $t \to \infty$, $x \to 0.01$. This means the bullet will not be able to penetrate the plate of material Q which is 0.1m thick, unlike what it has done to material P. Hence, the company's claim is valid. **Method 3** When $t = 0.0011204$, from (3), $\frac{\mathrm{d}x}{\mathrm{d}t} = 100\mathrm{e}^{-10^4(0.0011204)} = 0.00136 \approx 0$. In the same amount of time it takes for the bullet to penetrate the entire plate of material P to exit with a velocity of 80 ms^{-1}, the bullet that travels through the plate of material Q has its velocity reduced to almost 0 ms^{-1}. Hence, the company's claim is valid.

Source of Question: NYJC Prelim 9758/2025/01/Q12

9 Scientists are studying the growth of a newly discovered biomass that entered Earth's atmosphere together with an asteroid. A sample of 100 grams of the biomass was collected initially. It was found that on the next day, the sample grew to 110 grams. The scientists observed that the rate of growth of biomass was proportional to the amount of biomass present. The amount of biomass at time t days is denoted by B grams.

(a) Write down a differential equation relating B and t. [1]

(b) Solve the differential equation obtained in part **(a)**, expressing B in terms of t. [4]

(c) Sketch the graph of B against t, and explain what would happen if the biomass was left on its own. [2]

To limit the growth of the biomass, the scientist collected a second sample of 100 grams of the biomass and introduced a nutrient depletion serum to the sample the moment it was collected. The rate of growth of the biomass for this second sample is modelled by the differential equation

$$\frac{\mathrm{d}B}{\mathrm{d}t} = \frac{mB}{3+2t^2}$$

where m is a constant.

After 1 day, the second sample grew to 101 grams.

(d) For this second sample, find B in terms of t. [5]

(e) Determine the amount of the second biomass sample after a long time, leaving your answer to 5 decimal places. [1]

Solution:

(a)	$\frac{\mathrm{d}B}{\mathrm{d}t} = kB,$ where $k \in \mathbb{R}$.
(b)	$\frac{1}{B}\frac{\mathrm{d}B}{\mathrm{d}t} = k$ Integrating both sides w.r.t. t, $\int \frac{1}{B}\,\mathrm{d}B = \int k\,\mathrm{d}t$ $\ln B = kt + C,$ where C is an arbitrary constant (note that $B \geq 0$ since it refers to mass) $B = \mathrm{e}^{kt+C} = D\mathrm{e}^{kt}$ Given $B = 100$ when $t = 0$:

$100 = D$

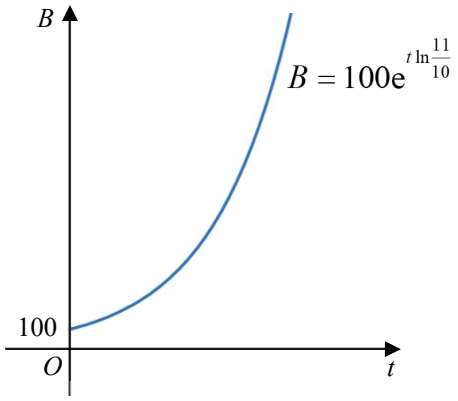
Given $B = 110$ when $t = 1$:

$110 = 100\mathrm{e}^{k}$

$k = \ln\frac{11}{10}$

$\therefore B = 100\mathrm{e}^{t\ln\frac{11}{10}}$ or $100\left(\frac{11}{10}\right)^{t}$

(c)



The biomass would grow indefinitely if it was left on its own.

(d)

$\frac{\mathrm{d}B}{\mathrm{d}t} = mB\frac{1}{3+2t^2}$, where $m \in \mathbb{R}$.

$\frac{1}{B}\frac{\mathrm{d}B}{\mathrm{d}t} = \frac{m}{3+2t^2}$

Integrating both sides w.r.t.t:

$$\int \frac{1}{B}\,\mathrm{d}B = \int \frac{m}{3+2t^2}\,\mathrm{d}t$$

$$= \frac{m}{2}\int \frac{1}{\frac{3}{2}+t^2}\,\mathrm{d}t$$

$$\ln B = \frac{m}{2}\sqrt{\frac{2}{3}}\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right) + C$$

$$B = \mathrm{e}^{\frac{m}{\sqrt{6}}\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right)+C}$$

$$= D\mathrm{e}^{\frac{m}{\sqrt{6}}\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right)}$$

Given $B = 100$ when $t = 0$:

$100 = D$

Given $B = 101$ when $t = 1$:

	$101 = 100\mathrm{e}^{\frac{m}{\sqrt{6}}\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)}$ $m = \left(\ln\frac{101}{100}\right)\frac{\sqrt{6}}{\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)} = 0.03559595$ $B = 100\mathrm{e}^{\frac{0.03559595}{\sqrt{6}}\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right)}$ $= 100\mathrm{e}^{0.014532\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right)}$ $= 100\mathrm{e}^{0.0145\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right)}$
(e)	As $t \to \infty$, $\tan^{-1}\left(\sqrt{\frac{2}{3}}t\right) \to \frac{\pi}{2}$ $\therefore B \to 100\mathrm{e}^{0.014532\left(\frac{\pi}{2}\right)} = 102.30893$ grams

Source of Question: TJC Prelim 9758/2025/01/Q12

10 **(a)** Show, by means of the substitution $w = x^2y$, that the differential equation

$$2y + x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^4y^2}{\sqrt{x^2+1}}$$

can be reduced to the form $\frac{\mathrm{d}w}{\mathrm{d}x} = \frac{w^2x}{\sqrt{x^2+1}}$. [2]

Hence find the general solution of the differential equation $2y + x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^4y^2}{\sqrt{x^2+1}}$, leaving your answer in the form $y = \mathrm{f}(x)$. [3]

(b) At a durian plantation, mature durians are susceptible to pests' infection. The spread of pests at the plantation resulted in durians being classified into two categories, either they are bad durians that have been infected by the pests or good durians that have not been infected. It is given that x denotes the number of bad durians, in thousands, in a fixed population size P, in thousands, where $P > 4$. The rate of spread of infection, $\frac{\mathrm{d}x}{\mathrm{d}t}$, where t represents the time taken in days, can be modelled as being proportional to the product

of the number of bad durians and good durians that have not been infected. It is given that $x = 4$ and $\frac{\mathrm{d}x}{\mathrm{d}t} = P - 4$ when $t = 0$.

(i) Write down the differential equation involving $\frac{\mathrm{d}x}{\mathrm{d}t}$, x and P [2]

(ii) Solve the differential equation in part **(i)** and find x in terms of P and t. [5]

(iii) By using an appropriate graph, explain in context, the long-term implications if no additional measures are carried out to control the spread of the pests' infection. [2]

Solution:

(a)

Differentiate $w = x^2 y$ with respect to x,

$$\frac{\mathrm{d}w}{\mathrm{d}x} = 2xy + x^2 \frac{\mathrm{d}y}{\mathrm{d}x} \quad \text{---------} (1)$$

Multiply x to given D.E., we have $2xy + x^2 \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^5 y^2}{\sqrt{x^2+1}}$ ------ (2)

Subst. (1) into (2) and replace $x^4 y^2$ by w^2 :

$$\frac{\mathrm{d}w}{\mathrm{d}x} = \frac{w^2 x}{\sqrt{x^2+1}}$$

$$\int \frac{1}{w^2}\,\mathrm{d}w = \int \frac{x}{\sqrt{x^2+1}}\mathrm{d}x$$

$$\int w^{-2}\,\mathrm{d}w = \frac{1}{2}\int 2x\left(x^2+1\right)^{-\frac{1}{2}}\mathrm{d}x$$

$$\frac{w^{-1}}{-1} = \frac{1}{2}\frac{\left(x^2+1\right)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + C \text{ where } C \text{ is arb. constant}$$

$$-\frac{1}{w} = \sqrt{x^2+1} + C$$

$$-\frac{1}{x^2 y} = \sqrt{x^2+1} + C$$

$$y = -\frac{1}{x^2\left(\sqrt{x^2+1} + C\right)}$$

(b)(i)

$\dfrac{\mathrm{d}x}{\mathrm{d}t} = kx\left(P-x\right)$ where $k>0$

Given that $x=4$ and $\dfrac{\mathrm{d}x}{\mathrm{d}t} = P-4$,

$P-4 = 4k\left(P-4\right)$

$k = \dfrac{1}{4}$

$\dfrac{\mathrm{d}x}{\mathrm{d}t} = \dfrac{1}{4}x\left(P-x\right)$

(ii)

$\dfrac{\mathrm{d}x}{\mathrm{d}t} = \dfrac{1}{4}x\left(P-x\right)$

$\displaystyle\int \frac{1}{x\left(P-x\right)}\,\mathrm{d}x = \int \frac{1}{4}\,\mathrm{d}x$

$\dfrac{1}{P}\displaystyle\int \frac{1}{x}+\frac{1}{P-x}\,\mathrm{d}x = \int \frac{1}{4}\,\mathrm{d}t$

$\ln\left|x\right| - \ln\left|P-x\right| = \dfrac{P}{4}t + C$ where C is arbitrary constant

$\ln\dfrac{x}{P-x} = \dfrac{P}{4}t + C$ where $P-x>0$ and $x>0$

$\dfrac{x}{P-x} = A\mathrm{e}^{\frac{P}{4}t}$ where $A = \mathrm{e}^{C}$

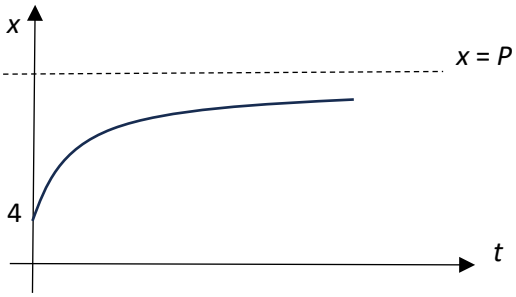
$\dfrac{P-x}{x} = \dfrac{1}{A}\mathrm{e}^{-\frac{P}{4}t}$

$\dfrac{P}{x}-1 = B\mathrm{e}^{-\frac{P}{4}t}$ where $B = \dfrac{1}{A}$

$x = \dfrac{P}{1+B\mathrm{e}^{-\frac{P}{4}t}}$

When $t=0$, $x=4$, $\dfrac{P}{4}-1 = B$

$x = \dfrac{P}{1+\left(\dfrac{P-4}{4}\right)\mathrm{e}^{-\frac{P}{4}t}} = \dfrac{4P}{4+\left(P-4\right)\mathrm{e}^{-\frac{P}{4}t}}$

(iii)	$x = \dfrac{4P}{4 + (P-4)\mathrm{e}^{-\frac{P}{4}t}}$  As $t \to \infty$, $\mathrm{e}^{-\frac{P}{4}t} \to 0$. $\therefore x \to P$ In the long term, all the durians in the plantation will be infected and become bad durians.